

What gets shown — and what gets overlooked?

We traced **8.6 million** logged article impressions from one week of Microsoft News back to the **65,238** articles in the dataset. What readers were shown did not mirror what was available: **entertainment** received **5.1×** its share of the catalogue, while **sports** — 30% of all articles — received **0.37×**.

30%

of articles were sports...

11%

...but only this share of what users were shown

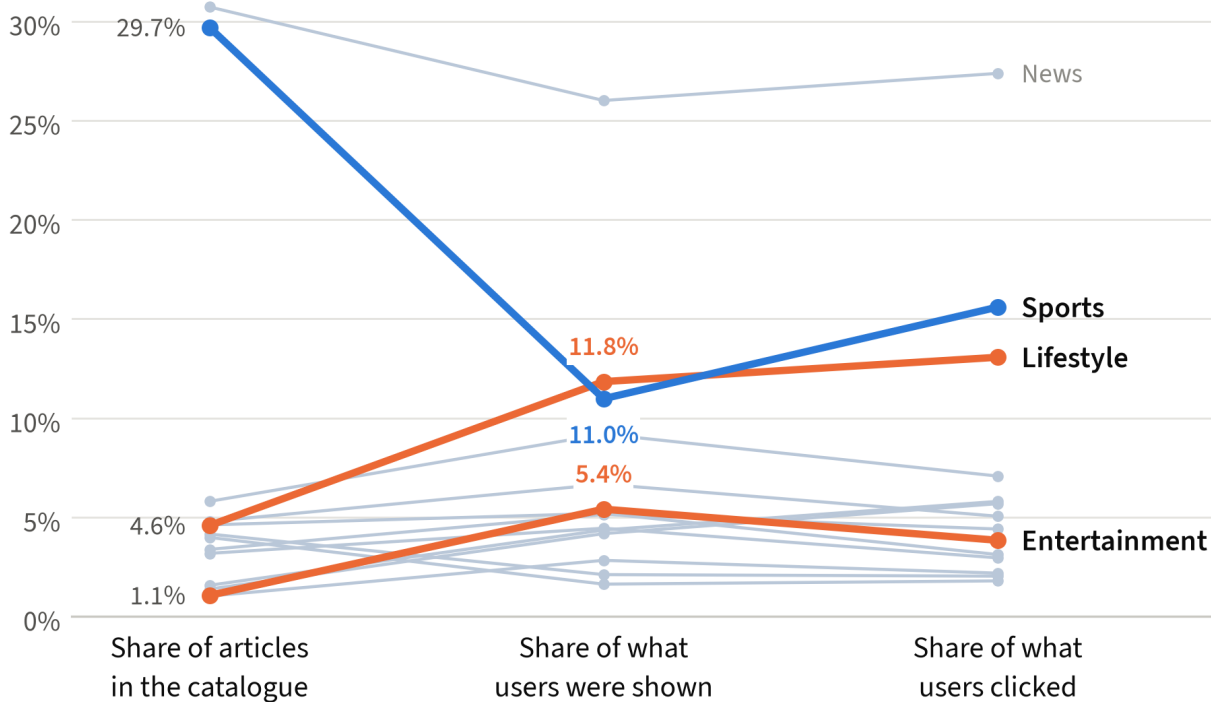
5.1×

Entertainment: exposure relative to its catalogue share

12%

Lifestyle: share of exposure, from 4.6% of articles

Share of each news category: in the catalogue → in what users were shown → in what they clicked



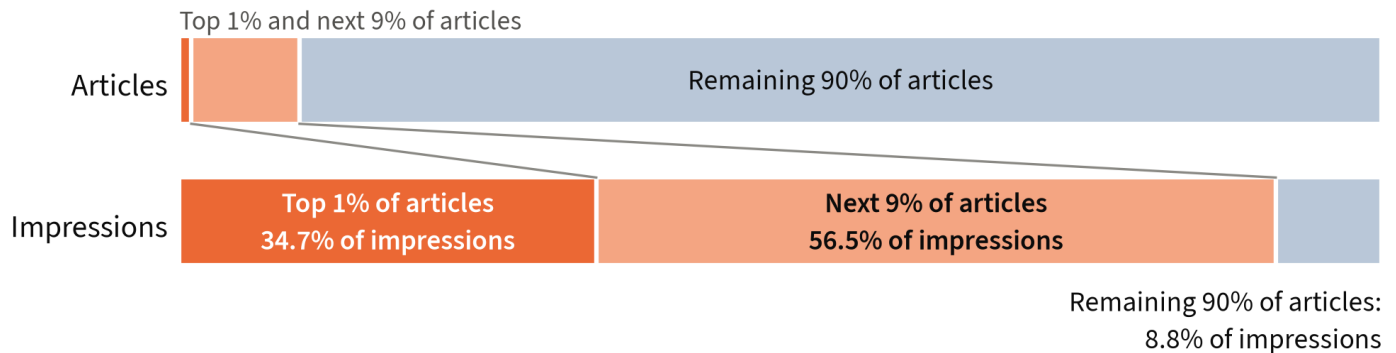
Each line is one of the 14 MIND categories with at least 50 articles. Orange: shown more than its catalogue share; blue: shown less; grey: other categories.

Sports was clicked most often when shown. Its click-through rate — clicks ÷ times shown — was 5.75%, the highest of any large category, yet it received the least exposure relative to its catalogue presence. Entertainment had one of the lowest click-through rates (2.88%) and the most amplified exposure.

What this is — and isn't. An *impression* means an article was logged as shown; it is a proxy for exposure, not proof of reading. The pattern describes the output of MSN's whole system — editors, page layout, recommendation and user navigation — not the recommender alone, and it does not show intent or unfairness.

35% of all exposure went to 1% of articles

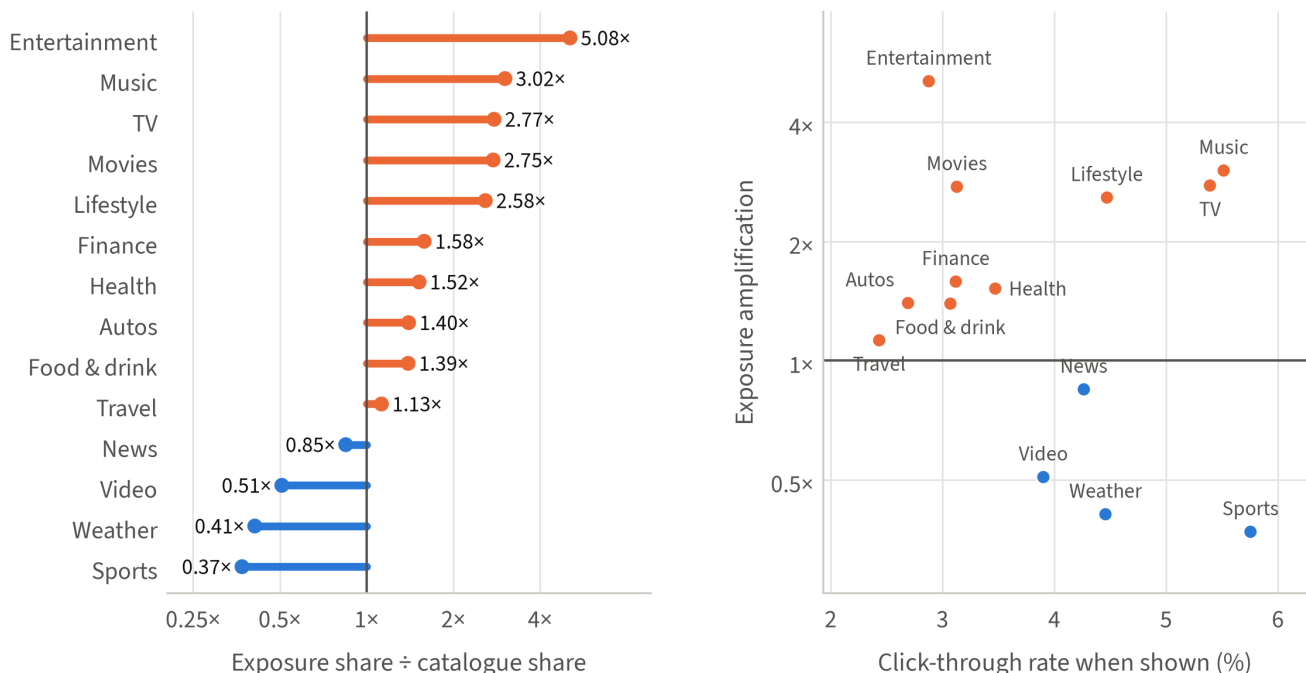
Of the 22,771 articles shown at least once, the most-shown 228 received 34.7% of all impressions and the top 10% received 91.2%. The bottom half of articles shared 0.4%.



Gini coefficient of impressions per article: 0.92 (0 = every article shown equally, 1 = one article shown everywhere). It stays between 0.89 and 0.93 within every single day, so the concentration is not just older and newer articles being mixed together.

Exposure tracks neither catalogue size nor clicks

Categories ranked by exposure relative to their catalogue share (left), and the same ratio against how often each category was clicked when shown (right).



Click-through rates are only comparable inside this dataset: every logged impression contains at least one click, so they describe engaged sessions. Four categories with fewer than 50 articles are excluded.

Does past clicking narrow what users are shown later?

PENDING — Notebook 02 results not found. Run Notebooks 02–04 on the MIND data, then rebuild.

Can a feed be relevant and diverse?

PENDING — Notebook 03 results not found. Run Notebooks 02–04 on the MIND data, then rebuild.

How we know — and what we can't conclude

Data. Microsoft News Dataset (MIND-small), Microsoft Research: 65,238 news articles with categories, and anonymised logs of 94,057 users — 230,117 impression logs (9–15 Nov 2019) containing 8,584,442 article impressions and 347,727 clicks.

Definitions. **Exposure** = an article logged as shown in an impression (a proxy, not proof of reading). **Engagement** = a click. **Catalogue share** = articles in a category ÷ all articles; **exposure share** = impressions ÷ all impressions; **CTR** = clicks ÷ impressions in that category.

Amplification. Exposure share ÷ catalogue share ($1\times$ = proportional). Because 65% of catalogue articles appear only in users' older click histories, we repeated it against articles actually shown at least once: the category ranking barely changes (rank correlation 0.97).

Concentration. Top-1%/10% shares and the Gini coefficient of impressions per article, also computed day by day.

Narrowing. Unbiased Simpson concentration (chance two items share a category), entropy, and a permutation test: users' favourite categories are shuffled within groups of similar activity and period to estimate exposure if it ignored history.

Recommenders. Popularity (train-week clicks), TF-IDF title similarity to the user's history, and favourite-category preference; MMR re-ranking with λ from 1 to 0. Evaluated on the dev day with nDCG@10, AUC, distinct categories, intra-list diversity, coverage and an amplification ratio. No dev clicks used as input.

Limitations

- Observational data: associations, not causes.
- One week of US Microsoft News; not the whole internet, other platforms, Canada, or other years.
- Users are anonymised; no demographic or location data exist, and none were inferred. Groups are behavioural.
- Impressions are an exposure proxy; they do not prove an article was seen or read.
- Every logged impression contains at least one click, so click-through rates describe engaged sessions and are comparable only within this dataset.
- Click histories pre-date the logs but have no timestamps and never update during the logged week.
- Observed exposure reflects MSN's whole ecosystem — editors, layout, recommendation, navigation — not one algorithm.
- Our recommenders are simulations on logged candidate lists; they do not reproduce MSN's proprietary system.
- Diversity is measured with 18 broad categories and title words; viewpoint and source diversity are not measured.

Sources. Wu, F. et al. (2020). MIND: A Large-scale Dataset for News Recommendation. *Proceedings of ACL 2020*, 3597–3606. Dataset: Microsoft News Dataset (MIND), Microsoft Research — <https://msnews.github.io/> (Microsoft Research License Terms). Carbonell, J. & Goldstein, J. (1998). The use of MMR, diversity-based reranking for reordering documents and producing summaries. *SIGIR '98*. **Reproducibility.** All code, notebooks and the result files behind every number: <https://github.com/suryalionael/information-diversity>. The dataset itself is not redistributed.